

A Hybrid LLM and Rule-Based Framework for Transparent Clinical Trial Eligibility Matching

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Abstract—Objective: Identifying eligible patients for oncology clinical trials remains a time-consuming task due to the large volume of unstructured clinical documentation and the complexity of eligibility criteria. Although Large Language Models (LLMs) have shown considerable promise for supporting this process, most existing approaches rely primarily on probabilistic reasoning, limiting transparency and reproducibility.

Methods: This paper presents a clinician-validated hybrid framework that combines LLM-based information extraction with deterministic rule-based eligibility assessment. Unstructured oncology records are transformed into structured patient profiles, while clinical trial eligibility criteria are extracted and converted into machine-interpretable representations. Patient eligibility is then assessed using deterministic evaluation whenever possible, with LLM-assisted reasoning reserved for criteria that cannot be formally represented. The framework was evaluated using three open-source 8B LLMs across five pipeline tasks.

Results: The best-performing models achieved 78.3% accuracy in patient profile extraction, 96.7% in trial structure identification, 92.3% in eligibility criteria extraction, up to 64.1% strict accuracy in eligibility criteria conversion, and 84.0% accuracy in patient-trial matching. Expert clinical evaluation further confirmed the reliability of the generated eligibility assessments, with justification quality scores exceeding 4.8 out of 5 for the best-performing models.

Conclusion: The results demonstrate that combining deterministic reasoning with lightweight LLMs provides a transparent, reproducible, and clinically reliable approach for supporting oncology clinical trial recruitment.

Index Terms—Large Language Models; Information Extraction; Clinical Trial Eligibility; Unstructured Clinical Data; Clinician-in-the-Loop; Semi-Deterministic Eligibility Evaluation; Structured Patient Data

eligibility from unstructured electronic health records. Clinicians must review lengthy clinical notes and compare them against extensive trial protocols, making patient recruitment a time-consuming and error-prone process.

Recent advances in Large Language Models (LLMs) have shown considerable promise for both clinical information extraction and patient-trial matching. However, most existing approaches rely predominantly on probabilistic reasoning or assume the availability of structured electronic health records, limiting transparency, reproducibility, and clinical auditability. Furthermore, information extraction and eligibility assessment are often treated as independent tasks rather than components of a unified clinical workflow.

This paper presents a clinician-validated hybrid framework that combines LLM-based information extraction with deterministic rule-based eligibility evaluation. The proposed pipeline transforms unstructured oncology notes into structured patient profiles, extracts and formalizes eligibility criteria from clinical trial protocols, and performs transparent patient-trial matching by combining logical rule execution with LLM-assisted reasoning only when deterministic evaluation is not possible.

The framework was evaluated using three open-source 8B-parameter LLMs across five pipeline tasks. The best-performing models achieved 92.3% accuracy for eligibility extraction, 78.3% for patient profile extraction, and up to 84.0% in patient-trial matching while producing clinician-rated justification scores of 4.93/5. These results demonstrate that combining deterministic reasoning with lightweight LLMs can support transparent and clinically reliable decision support for oncology trial recruitment.

The table of Statement of Significance table is present below in Table I.

INTRODUCTION

Oncology clinical trials remain essential for advancing cancer treatment, yet a substantial proportion of eligible patients are never enrolled due to the complexity of manually assessing

TABLE I: Statement of Significance

Problem or Issue	Oncology departments rely heavily on unstructured clinical documentation, making patient screening for clinical trials time-consuming, inconsistent, difficult to scale, and error-prone, leading to increased administrative burden and missed enrollment opportunities.
What is Already Known	Previous studies have shown that LLMs can extract clinical information or assist with patient-trial matching. However, many approaches rely solely on probabilistic model outputs, provide limited clinician validation, or do not separate information extraction from eligibility evaluation.
What this Paper Adds	This work proposes a clinician-validated pipeline that transforms free-text oncology diaries into structured patient profiles, converts trial criteria into logical rules, and performs semi-deterministic patient-trial matching through deterministic evaluation of structured criteria complemented by LLM-assisted reasoning for schema-independent requirements.
Who would Benefit from the New Knowledge	Clinical researchers, oncologists, hospitals adopting AI-assisted workflows, and researchers developing trustworthy LLM-based clinical decision support systems.

RELATED WORK

Recent advances in Large Language Models (LLMs) have significantly improved clinical natural language processing, particularly for extracting structured information from unstructured electronic health records [11], [12]. Several studies have demonstrated that instruction-tuned LLMs can identify diagnoses, treatments, biomarkers, and other clinically relevant entities with high accuracy [1], [3], [9]. Nevertheless, these approaches typically focus on information extraction as an isolated task and rarely integrate the extracted information into complete clinical decision-support workflows [9], [11].

Clinical trial matching has also benefited from recent advances in LLMs. Existing approaches commonly rely on semantic similarity, retrieval-augmented generation, or end-to-end probabilistic reasoning to assess patient eligibility [4], [8], [12]. While these methods achieve encouraging results, they often lack transparency and make it difficult for clinicians to verify how individual eligibility decisions are reached [4], [12]. Furthermore, many assume structured electronic health records, limiting their applicability in routine clinical practice [4], [6].

With these gaps in mind, the proposed framework combines LLM-based information extraction with deterministic rule evaluation and clinician validation throughout the workflow. Rather than relying exclusively on probabilistic reasoning, eligibility assessment is performed deterministically whenever possible, with LLM-assisted reasoning reserved for criteria that cannot be formally represented. This hybrid design builds upon previous hybrid approaches [3], [4] while explicitly separating structured information extraction from eligibility reasoning, improving transparency, traceability, and clinician oversight throughout the complete patient-trial matching workflow.

PROPOSED FRAMEWORK

The proposed framework aims to reduce the workload associated with patient screening by leveraging LLMs to extract clinically relevant information from free-text oncology records and support transparent clinical trial eligibility assessment. Rather than relying exclusively on probabilistic model decisions, the framework incorporates clinician validation throughout the pipeline, ensuring that all structured patient profiles and eligibility criteria are reviewed before being used for patient matching.

The workflow is organized into three interconnected phases:

- Phase 1: Hyper-Relevant Parameter Extraction;
- Phase 2: Extraction and Conversion of Clinical Criteria;
- Phase 3: Semi-Deterministic Trial-Patient Matching.

Hyper-Relevant Parameter Extraction

The first phase transforms free-text oncology clinical notes into normalized structured patient profiles. Clinical diaries selected by the clinician are first pre-processed to remove irrelevant content before being provided to the LLM together with a task-specific prompt.

The model extracts clinically relevant parameters according to a predefined patient schema while simultaneously normalizing diagnoses using a stakeholder-defined mapping. The generated patient profile is then presented to the clinician for validation and possible corrections. Once approved, the structured profile is stored and becomes available for the remaining stages of the pipeline.

A formalization of this workflow can be seen on the Figure 1 below.

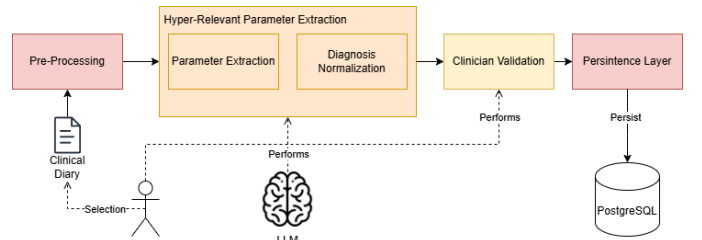


Fig. 1: Hyper-Relevant Parameter Extraction

Extraction and Conversion of Clinical Criteria

The second phase converts clinical trial eligibility requirements into machine-interpretable representations suitable for automated evaluation. The process begins by pre-processing the clinical trial document, removing sections unrelated to eligibility assessment. The LLM then identifies the trial structure, including the existence of different cohorts, providing additional context for the extraction stage.

Using this information, the model extracts inclusion and exclusion criteria while preserving their association with the

corresponding cohort whenever applicable. Individual criteria are extracted whenever possible, although compound criteria are preserved when necessary to maintain their clinical meaning. The extracted criteria are subsequently validated by the clinician before being translated into structured logical expressions. These logical representations are again reviewed by the clinician to ensure that they accurately capture the original intent of the eligibility criteria before becoming available for patient matching.

A formalization of this workflow can be seen on the Figure 2 below.

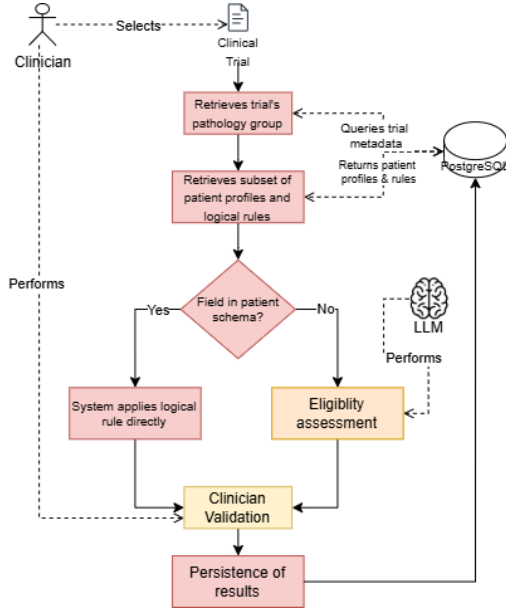


Fig. 2: Extraction and Conversion of Clinical Criteria Workflow

Semi-Deterministic Trial-Patient Matching

The final phase performs patient-trial matching using the validated artifacts produced by the previous stages. Structured patient profiles generated during Phase 1 are combined with the validated logical eligibility criteria obtained during Phase 2 to identify potentially eligible patients for a selected clinical trial.

After selecting a trial, the system retrieves a subset of patient profiles based on the pathology group defined during trial registration, reducing unnecessary comparisons and LLM calls. Each patient is then evaluated against every eligibility criterion using one of two complementary assessment strategies:

- *Deterministic Assessment* - if the criterion references a field available in the patient profile schema, the corresponding logical rule is evaluated directly against the structured patient data;
- *LLM-Assisted Assessment* - if the criterion references information that is not represented within the structured schema, the LLM receives the patient profile, clinical

diary, laboratory results, and the corresponding criterion to determine eligibility. In addition to the assessment result the model provides a justification to support expert review.

Although different evaluation mechanisms are employed, both strategies produce the same output: a criterion-level eligibility assessment. The complete set of assessments is then presented to the clinician for review, allowing any automatically generated decision to be confirmed or modified. Finally, all criterion evaluations, manual corrections, and the resulting eligibility decision are stored for traceability and future auditing.

A formalization of this workflow can be seen on the Figure 3 below.

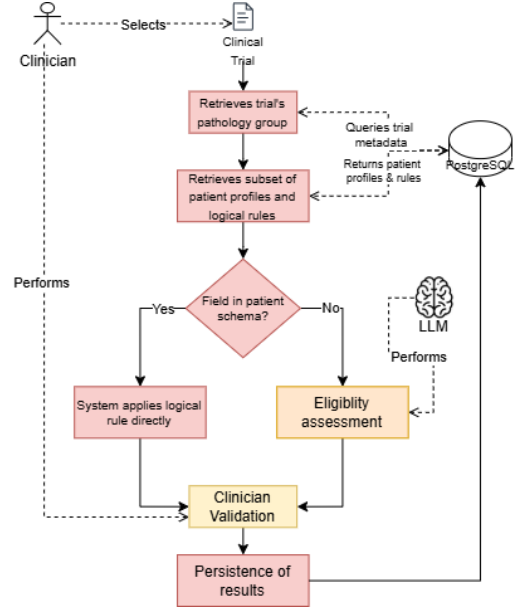


Fig. 3: Semi-Deterministic Trial-Patient Matching

EXPERIMENTAL EVALUATION

This section describes the experimental setup used to evaluate the proposed framework, including the datasets, evaluated models, and the obtained results.

Datasets

The proposed framework was evaluated using two datasets, together with manually curated gold standards used as reference for all evaluation tasks. All data used in this study were fully anonymized: no identifiable patient information was used, and every synthetic diary contains no real clinical content. A quantitative summary of the normalized datasets is provided in Table II.

The first dataset consists of 30 synthetic free-text oncology clinical diaries, generated with the open-source gpt-oss-120b model and grounded on four representative

real clinical diaries provided by the clinical stakeholder, which served as references for writing style, structure, and clinical content. Generation followed two sequential stages designed to reproduce the variability of real-world documentation:

- an idealized generation stage, in which the LLM produced a clinically coherent Portuguese oncology note consistent across diagnosis, staging, treatments, imaging findings, ECOG performance status, comorbidities, and disease evolution; and a
- noise and artifact insertion stage, in which the note was rewritten to resemble routine hospital records by randomly assigning a length mode (short, medium, or long) and a style mode (narrative-dominant, telegraphic hospital style, exam-and-imaging focused, toxicity focused, or psychosocial emphasis), introducing irregular formatting, heterogeneous abbreviations, minor spelling imperfections, structural asymmetries, and administrative workflow notes.

Authentic clinical documentation was used in the prompts strictly as a stylistic reference, with explicit instructions not to reproduce original clinical content, and an experienced oncologist reviewed all diaries for clinical plausibility and consistency. After normalization, the resulting corpus totaled 64,953 tokens across the 30 diaries (2,165 tokens on average).

The second dataset comprises 30 clinical trials collected from *ClinicalTrials.gov*, selected to maximize clinical diversity and minimize selection bias. The selected studies cover a diverse range of tumour types (including breast, lung, pancreatic, colorectal, and melanoma, among others), disease stages, molecular profiles, and therapeutic settings, and include both cohort-structured and non-cohort designs with substantial variability in eligibility complexity, from simple demographic or diagnostic constraints to specialized molecular biomarkers and prior treatment histories. After normalization, the corpus totaled 120,786 tokens across the 30 protocols (4,026 tokens on average), comprising 892 individual eligibility criteria (480 inclusion and 412 exclusion), averaging approximately 30 criteria per trial.

Manually curated gold standards were produced for three of the five core evaluation tasks. For parameter extraction, all fields defined in the patient profile schema were manually annotated within the 30 diaries to enable deterministic comparison with model outputs. For trial structure identification, the organization of eligibility requirements and the presence of cohorts were manually mapped following the system’s formal representation. For criteria extraction, all 892 inclusion and exclusion criteria were manually extracted as individual requirements to assess completeness of the automated extraction. For the remaining two tasks, no gold standard was defined in advance; instead, model outputs were assessed directly by manual review. For criteria conversion, the correctness of each generated logical formalization was manually assessed against the original eligibility criterion. For patient-trial matching, each LLM-generated eligibility decision was

manually assessed together with its accompanying justification to determine whether the decision was correct.

TABLE II: Quantitative summary of the normalized datasets used in this study.

Dataset Component	Total	Average per Document
Clinical Diaries (tokens)	64,953	2,165
Clinical Trials (tokens)	120,786	4,026
Inclusion Criteria	480	16
Exclusion Criteria	412	14
Total Criteria	892	30

Models

Three open-source Large Language Models from different model families were evaluated:

- Qwen3 8B Instruct;
- Llama 3.1 8B Instruct;
- DeepSeek-R1 8B.

All experiments were performed using deterministic inference settings to ensure reproducibility and minimize output variability. No fine-tuning was performed, and all models were evaluated under a zero-shot prompting strategy.

Experimental Setup

The proposed framework comprises three LLM-assisted pipelines: Hyper-Relevant Parameter Extraction, Extraction and Conversion of Clinical Criteria, and Semi-Deterministic Patient–Trial Matching. Each LLM-dependent component was evaluated independently to isolate potential error sources and clarify each stage’s contribution to overall system performance, with particular emphasis on accuracy, reliability, interpretability, and suitability for operation under clinician supervision. All models were evaluated under identical conditions, using version-controlled prompts and fixed configuration parameters throughout.

Evaluation scenarios. Three scenarios were defined, each targeting a different pipeline. Scenario 1 (Clinical Diary Processing) evaluates the Parameter Extraction pipeline: each model receives a clinical diary and must extract and normalize the information required to populate the patient profile schema, assessing its ability to identify clinically relevant information in heterogeneous free text while complying with the target structure. Scenario 2 (Clinical Trial Processing) evaluates the Criteria Extraction and Conversion pipeline: each model receives a trial protocol and must identify the eligibility structure (including cohort detection), extract inclusion/exclusion criteria, and convert them into machine-interpretable logical representations, assessing whether clinical meaning is preserved through this transformation. Scenario 3 (Patient–Trial Matching) evaluates the LLM-assisted component of the matching pipeline, used when a criterion cannot be resolved by deterministic rules alone (due to incomplete formalization or schema-independent requirements); the model receives the patient profile, original diary, relevant laboratory data, and

the criterion as a logical rule, and must decide whether the criterion is satisfied and justify the decision, assessing both eligibility-assessment reliability and the transparency of the generated justification under constrained clinical context.

Evaluation metrics. Metrics were tailored to each task. For parameter extraction: field-level accuracy against a manually annotated gold standard, missing-field rate, and schema compliance rate. For trial structure identification: cohort detection accuracy and expert assessment of structure correctness. For criteria extraction: expert assessment of whether extracted criteria faithfully represent the original trial requirements. For criteria conversion: manual assessment of whether the generated logical representation accurately reflects the criterion’s clinical meaning. For patient-trial matching: eligibility assessment accuracy relative to the patient’s reference documentation, and justification quality, rated by experts on a five-point Likert scale.

Results

The proposed framework was evaluated across five inter-dependent tasks spanning structured information extraction, logical formalization of eligibility criteria, and patient-trial matching. With the exception of the final matching stage, every task was scored under a three-way scheme (correct, partial, wrong). We report both the strict accuracy (correct outputs only) and a partial-aware accuracy that additionally credits partially correct outputs. Table III lists the best-performing model for each task. Rather than discussing each task in isolation, we organize the analysis around three questions: how much accuracy was achieved at each stage, what kinds of errors recurred across stages, and how expert clinicians judged the quality and trustworthiness of model outputs.

TABLE III: Pipeline Performance Across Tasks

Task	Best Model	Accuracy
Patient Profile Extraction	Qwen3	78.3%/83.3%
Trial Structure Identification	DeepSeek	96.7%/96.7%
Criteria Extraction	Qwen3	92.3%/95.9%
Criteria Extraction - cohort specific	Qwen3	96.3%/96.3%
Criteria Conversion	DeepSeek	64.1%/73.4%
Patient Matching	Qwen3	84.0%/84.0%

Accuracy Across the Pipeline: In *patient profile extraction*, all models produced schema-valid JSON in 100% of cases. Qwen3 obtained the best result (78.3% strict / 83.3% partial accuracy, 11.6% missing-field rate), followed by DeepSeek (74.3%/80.5%, 13.3% missing) and Llama3.1 (62.3%/70.3%, 21.7% missing), these results can be seen on Figure 4.

Trial structure identification showed the widest spread of any task: DeepSeek reached 96.7% strict accuracy, Qwen3 60% (75% partial), and Llama3.1 only 25%, these results can be seen on Figure 4.

Criteria extraction was the strongest stage overall. On individual inclusion/exclusion criteria, 92.3%/95.9%, DeepSeek 83.7%/88.2%, and Llama3.1 71.9%/75.8%. On *cohort attribution of criteria*, Qwen3 again led with 96.3%, while DeepSeek

and Llama3.1 both reached 66.7%, these results can be seen on Figure 4.

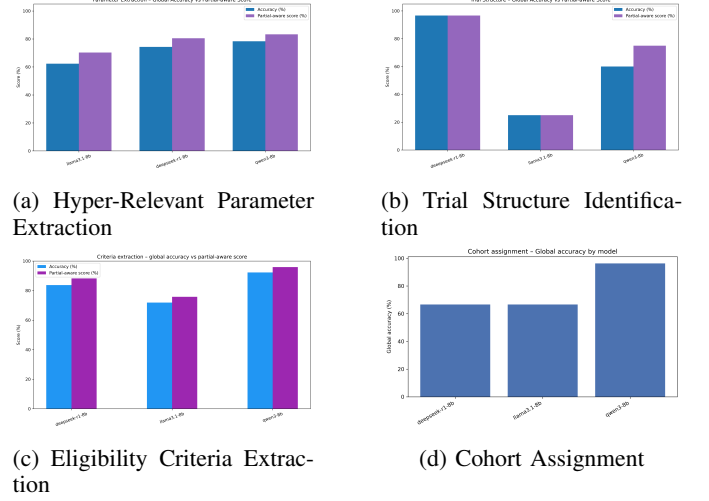


Fig. 4: Accuracy across the structured extraction stages of the pipeline.

Criteria conversion was the pipeline’s bottleneck, see results on Figure 5. Among the 8B-parameter models, Qwen3 achieved the highest accuracy 65.1%/74.4%) but also the most outright errors, Llama3.1 was comparable (62.5%/73.5%), DeepSeek had the lowest strict score (52.2%) but the highest increase in partial score (71.2%). Testing a larger model, gpt-oss-120b reached 65.1% strict and 80.9% partial accuracy with markedly fewer incorrect conversion than any 8B model.

Patient-trial matching recovered sharply despite the weaker conversion scores, see results on Figure 5. Qwen3 reached 84.0% accuracy, DeepSeek 83.0%, and Llama3.1 78.0%.

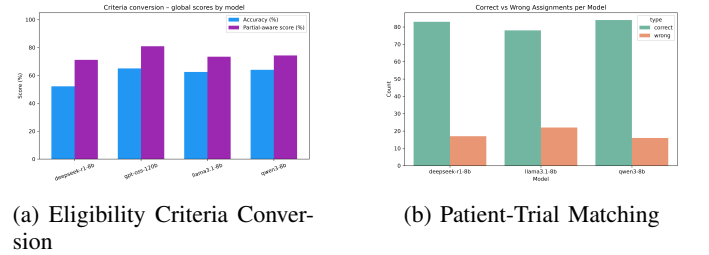


Fig. 5: Accuracy of criteria conversion, and final patient-trial matching.

Failure Patterns Across Tasks: Looking across stages rather than within a single task reveals three recurring failure modes.

The first concerns temporal and hierarchical complexity in patient data. In profile extraction, errors concentrated in diagnosis normalization, hierarchical treatment history, full TNM staging, and date-related fields, while simple explicit fields such as Gender and ECOG-PS were solved perfectly by every model. This indicates that errors stemmed from structural and normalization demands rather than from difficulty recognizing individual clinical entities

The second concerns cohort and criteria level representation. In trial structure identification, weaker models mainly confused descriptive patient groups with formally defined cohorts and struggled with schema compliance, expert review noted that even correctly identified cohorts did not always preserve the defining information of the original trial. In criteria extraction, deviations were again largely representational, omission or inconsistent fragmentation of compound criteria, rather than misreadings of clinical intent, with Llama3.1 fragmenting compound criteria most frequently and producing the most outright incorrect criteria.

The third concerns nested logic and schema mapping during criteria conversion. Across all models, criteria combining multiple AND/OR conditions were frequently under-represented, with only the most prominent condition captured and secondary constraints dropped. A second recurring issue was the misassignment of treatment-related constraints to newly invented, non-schema fields instead of existing patient-schema fields, which unnecessarily pushed evaluation downstream onto LLM-based reasoning. Notably, most of the residual errors made by the larger gpt-oss-120b model were of this schema-normalization type rather than failures to capture logical structure, suggesting that scaling model capacity mitigates, but does not eliminate, this problem.

Finally, at the matching stage, errors did not stem from partial rule satisfaction but from outright misinterpretation of clinical evidence, most visibly in Llama3.1, which occasionally over-interpreted uncertainty, for instance rejecting a patient despite adequate laboratory values or drawing conclusions from incomplete documentation.

Expert Qualitative Assessment: Beyond accuracy scores, clinician review assessed whether model outputs were clinically meaningful and defensible.

At the trial structure identification stage, expert review noted that correctly identifying the presence of a cohort structure did not necessarily guarantee that the generated representation preserved the defining clinical information of the original trial, indicating that structural correctness and clinical completeness are not always equivalent and should be assessed separately.

At the extraction stages, experts confirmed that Qwen3 consistently preserved the clinical meaning of extracted criteria even when restructuring compound statements, and that DeepSeek maintained high semantic fidelity without hallucinating content. Llama3.1's behavior was more variable these findings reinforce that clinician-in-the-loop validation remains necessary when transforming free-text eligibility language into computation representations, even for the best-performing models.

At the matching stage, expert-assigned justification scores tracked accuracy closely. Qwen3 scored highest (4.93/5), consistently grounding decision in explicit patient data and defaulting to conservative reasoning when information was missing, DeepSeek scored 4.8/7, reliably distinguishing be-

tween information that was genuinely absent and information that was merely unavailable in the record, without making unsupported assumption, Llama3.1 scored 4.2/5, performing well when evidence was explicit but occasionally over-interpreting missing or incomplete data. Taken together, these results indicate that reliability in patient-trial matching depends not only on raw accuracy but on the transparency and evidential grounding of the underlying justification, a property that is essential for trustworthy clinical decision support and that purely numeric accuracy scores do not capture.

DISCUSSION

The experimental results demonstrate that lightweight open-source LLMs are capable of supporting multiple stages of a clinical trial matching workflow, although performance varies according to task complexity. Across all evaluated models, structured extraction tasks consistently achieved the highest accuracies, while transforming natural-language eligibility criteria into formal logical representations remained the principal bottleneck of the pipeline.

The results also highlight that no single model consistently outperformed the others across all tasks. Qwen3 achieved the best performance in patient profile and eligibility criteria extraction, whereas DeepSeek excelled in trial structure identification and demonstrated robust reasoning during patient-trial matching. These observations suggest that decomposing the workflow into specialized components enables each task to benefit from the strengths of different models, rather than relying on a single end-to-end solution.

Finally, the proposed hybrid strategy combines deterministic rule evaluation with LLM-assisted reasoning only when structured information is insufficient. Together with clinician validation at every stage, this design improves transparency and traceability while reducing dependence on purely probabilistic model decisions, characteristics that are essential for trustworthy clinical decision-support systems.

CONCLUSION

This paper presented a clinician-validated hybrid framework for semi-deterministic patient-trial matching from unstructured oncology clinical documentation. The proposed methodology combines LLM-based information extraction, logical formalization of eligibility criteria, deterministic rule evaluation, and LLM-assisted reasoning to support transparent clinical trial recruitment.

The experimental evaluation showed that lightweight open-source LLMs are capable of performing clinically relevant tasks across the proposed pipeline. Structured extraction achieved the strongest results, with Qwen3 reaching 78.3% strict accuracy for patient profile extraction and 92.3% for eligibility criteria extraction, while patient-trial matching achieved up to 84.0% accuracy. In contrast, logical formal-

ization remained the most challenging stage, highlighting an important direction for future research.

Overall, the proposed framework demonstrates that combining structured representations, deterministic reasoning, and clinician supervision provides a practical and explainable approach for supporting oncology clinical trial recruitment. Future work will focus on evaluating larger language models, improving logical criteria formalization, and validating the framework using larger collections of real-world clinical documentation.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author(s) used Large Language Models, such as ChatGPT, or similar tools for the refining of the work’s contents. The author(s) reviewed and edited the output as needed and take full responsibility for the content of the published article.

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